

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology.
(BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

I Thanigavel S. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thanigavel

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5.
Problem Understanding	20%	1.5	1.5	1.5	1.5	1.5
Method Selection	20%	1.5	1.5	1.5	1.5	1.5
Solution Process	25%	2	2	2	2	2
Accuracy of Calculation	20%	1	1	1	1	1
Interpretation of Results	15%	1/2	1/2	1/2	1/2	1/2.

Total Marks 38 /40

[Signature]
Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department.

Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		accurately			
		Selects most appropriate method/ formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Method Selection	20%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Solution Process	25%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Accuracy of Calculation	20%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Interpretation of Results	15%				

1. Problem Analysis:

The administrator has divided $192.168.10.0/24$ into for 126 Subnets.

$A/26$ Subnet Provides.

→ Total addresses. = 64.

→ Usable hosts = 62.

* Use VLSM instead of FLSM

Possible allocations:

* 50 hosts → $/26$.

* 25 hosts → $/27$.

* 15 hosts → $/27$.

* 10 hosts → $/28$.

Analysis:

Although every department can accommodate its required hosts, the design wastes many IP addresses.

Example:

* Department 4 required only 10 hosts but receives 52 usable addresses

* Wasted addresses = 52.

Department requirements:

Department	Hosts required	/26 Supports.
Dept 1	50	Yes.
Dept 2	25	Yes.
Dept 3	15	Yes.
Dept 4	10	Yes.

Conclusion:

The administrator's design is functional but inefficient. VLSM should be adopted because it minimizes IP addresses wastage and allows easier future expansion.

2. Problem Analysis.

PC1:

* 192.168.5.62 /26.

PC2:

* 192.168.5.65 /26.

for a /26 subnet:

Subnet ranges are:

* 192.168.5.0 - 63

* 192.168.5.67 - 127.

* 192.168.5.128 - 191

* 192.168.5.192 - 255.

Analysis.

PC 1 (192.168.5.62)

* Network = 192.168.128 - 191.

* 192.168.5.128 - 191.

* 192.168.5.192 - 255.

Analysis:

PC1 (192.168.5.62)

* Network = 192.168.5.0.

* Broadcast = 192.168.5.63.

PC2 (192.168.5.65)

* Network = 192.168.5.64.

* Broadcast = 192.168.5.127.

The system belong to different subnets.

Conclusion:

The PCs cannot communicate directly because they are in different subnetworks, communication is possible only through a router (or) Layer-3 devices.

3. Problem Analysis:

Network

172.16.20.0 /24.

Chosen Subnet mask:

/27.

A/27 Subnet provides.

- * 32 total addresses.

- * 30 usable hosts.

One department requires 40 hosts.

Therefore, the chosen Subnet mask is not suitable.

A better choice is:

- * Department requiring 40 hosts $\rightarrow$ /26 (62 usable hosts)

- * Remaining departments $\rightarrow$ /27.

Conclusion:

The Subnet mask does not satisfy all department requirement, using VLSM with different subnet sizes provides better utilization and scalability.

Problem Analysis:

Router: 10.1.95/27

Host : 10.1.1.100 /27

For /27

Subnet ranges are:

* 10.1.1.64 - 95.

* 10.1.1.96 - 127.

Analysis:

Router IP = 10.1.1.95.

This is the broadcast address of Subnet 10.1.1.64/27

Broadcast address cannot be assigned to devices.

Host IP = 10.1.1.100. it belongs to Subnet 10.1.1.96/27.

Thus;

* Router configuration is invalid.

* Router and hosts are also in different subnets.

Conclusions:

Communication is not possible, the router interface should be assigned a valid host IP (for example, 10.1.1.97/27) before communication can occur.

5. Problem Analysis:

IPv6 Address:

2001: 0DB8 : 1234 :: 0000 : 0000 : ABCD : 0000 : 0015 :

Expanded form:

2001: 0DB8: 1234: 0000 : 0000: ABCD : 0000 : 0015 :

Network prefix (164):

200001: 0DB8 : 1234 : 0000 :: /67.

Host identifier:

0000: ABCD : 0000 : 0015.

IPv6 Advantage :- IPv6. Provides an enormous address space (2^{128} addresses), eliminating IPv4 addresses exhaustion and reducing the need for Network Address Translation (NAT).

Conclusion:

The address is correctly represented after compression and expansion. The 164 prefix identifies the network, while the remaining 64 bits represent the interface identifiers. IPv6 is better suited for modern enterprise networks because of scalability, simplified routing and vast address space.

vision:

Using separate subnets reduces broadcast traffic, improves security and allows both LANs to communicate efficiently through the layer-3 router using their configured gateways.